



B. E–VII Sem- Project Seminar (2026-27)

Department of Electronics and Communication Engineering

Vasavi College of Engineering (Autonomous)

Ibrahimbagh, Hyderabad - 500031

ABSTRACT

Student 1 : K. Deveswar 1602-23-735-136 } **Supervisor:** Mr. V Krishna Mohan

Student 2 : P. Mourya Sai 1602-23-735-154 ‡ **Designation:** Assistant Professor

Student 3 : E. Venkata Harsha Vardhan 1602-23-735-189 } ‡

Title of the Project:

Development of AUTOSAR Mixed Signal Timing Extension for Automotive Embedded System Modeling

ABSTRACT:

Modern automotive embedded systems increasingly use multicore processors to meet computational and real-time requirements. However, cross-core system calls, synchronization mechanisms, and inter-processor interrupts (IPIs) introduce complex interactions that make static analysis and timing-aware optimization challenging. This project first focuses on reproducing and validating the Multi-core Static State Exploration (MultiSSE) approach proposed by Entrup et al. for multicore AUTOSAR systems. MultiSSE analyzes cross-core interactions and constructs a Multi-core State Transition Graph (MSTG) using program states and synchronization points. BCET and WCET information is used to identify infeasible state combinations and enable optimization of unnecessary synchronization operations and IPIs. The reproduced approach will be evaluated on a Raspberry Pi 5 and compared with the methodology and results reported in the reference work.

In the subsequent phase, the validated framework will be extended to incorporate mixed-signal timing information from automotive sensor interfaces. Sensor circuits will be simulated using Cadence Virtuoso/ADE, with timing characteristics extracted from transient waveforms and processed in MATLAB. These characteristics will be mapped to relevant AUTOSAR timing concepts and integrated using MATLAB/Simulink AUTOSAR Blockset, establishing a connection between mixed-signal circuit timing and AUTOSAR software timing analysis.

Keywords

AUTOSAR, Multicore Systems, Real-Time Systems, Static Analysis, MultiSSE, MSTG, Timing Analysis, BCET, WCET, Lock Elision, IPI Avoidance, Syscall Cost Model, Mixed-Signal Systems, Sensor Timing

Tools

ARA
LLVM
Trampoline RTOS
dOSEK
C/C++ and Python
Raspberry Pi 5
ARM Performance Monitoring Unit (PMU)
MATLAB / Simulink
AUTOSAR Blockset
Cadence Virtuoso / ADE

References

1. G. Entrup, A. Kässens, B. Fiedler, and D. Lohmann, "Applied static analysis and specialization of cross-core syscalls for multi-core AUTOSAR OS," *Real-Time Systems*, vol. 60, pp. 491-533, 2024, doi: 10.1007/s11241-024-09429-1.
 2. G. Entrup, A. Kässens, B. Fiedler, and D. Lohmann, "Applied static analysis and specialization of cross-core syscalls for multi-core AUTOSAR OS," in *Proceedings of the 29th IEEE International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA)*, 2023.
- This is the earlier conference version extended by the 2024 journal paper.

3. C. Lattner and V. Adve, "LLVM: A Compilation Framework for Lifelong Program Analysis & Transformation," in Proceedings of the 2004 International Symposium on Code Generation and Optimization (CGO-04), 2004.
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5. AUTOSAR, "Specification of Operating System," AUTOSAR Classic Platform, Automotive Open System Architecture.
6. AUTOSAR, "Specification of Timing Extensions," AUTOSAR Classic Platform, Document ID 411.
7. AUTOSAR, "Requirements on Timing Extensions," AUTOSAR Foundation, Document ID 410.

IMPLEMENTATION PLAN:

1. Multicore AUTOSAR System Setup: Configure a statically defined multicore AUTOSAR system with tasks, synchronization mechanisms, and cross-core system calls using the selected AUTOSAR and compiler infrastructure.
2. Static State Exploration: Analyze application control flow and core-local states to identify synchronization points and possible cross-core interactions, and construct the Multi-core State Transition Graph (MSTG).
3. Timing-Aware Analysis and Optimization: Incorporate BCET and WCET information to identify infeasible state combinations and support optimization of cross-core interactions through lock elision and IPI avoidance.
4. System Evaluation: Execute the resulting multicore system on a Raspberry Pi 5 and evaluate its execution behavior using ARM Performance Monitoring Unit (PMU) measurements.
5. Mixed-Signal Timing Extension: Extend the timing-analysis framework by extracting sensor timing characteristics from Cadence Virtuoso/ADE simulations, processing the extracted data using MATLAB/Simulink, and mapping the resulting timing information to relevant AUTOSAR Timing Extension concepts using AUTOSAR Blockset.

DESIGN METHODOLOGY:

The proposed methodology models a statically configured multicore AUTOSAR system through its application control flow, atomic basic blocks (ABBs), and core-local execution states. Cross-core system calls are identified as interaction points between processor cores, and synchronization points (SPs) are established to capture the relative execution states of the affected cores. These states are used to construct the Multi-core State Transition Graph (MSTG), representing the feasible cross-core interactions while avoiding exhaustive enumeration of the complete multicore state space. BCET and WCET information is incorporated as timing constraints during state exploration to eliminate infeasible state combinations. The resulting state information supports specialization of cross-core system calls through lock elision and IPI avoidance.

The methodology is further extended to represent timing characteristics originating from mixed-signal automotive sensor interfaces. Sensor circuits are modeled and simulated using Cadence Virtuoso/ADE, with relevant timing characteristics extracted from transient waveforms. The extracted timing information is processed using MATLAB/Simulink and mapped to corresponding AUTOSAR Timing Extension concepts through AUTOSAR Blockset, establishing a relationship between sensor-level timing behavior and software-level timing analysis.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3				2						3			
CO2		2			3							3			
CO3	2	2		3			3					3			
CO4								3	2	3					
CO5									3			3			

Note: CO1 & CO2 must be mapped with one of the relevant PSOs based on the domain of the project with 3

CO3: can be mapped to appropriate PSO with level 2

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Supervisor

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3.
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Student(s)